

CURRENT AFFAIRS & SCIENCE FOR UPSC CSE 2026 (INTEGRATED NOTEBOOK)

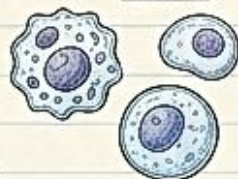
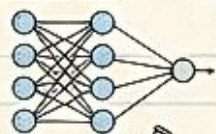
1. HANTAVIRUS & M.V. HONDIUS EXPEDITION

- **CRUISE SHIP & VOYAGE:** M.V. Hondius, traveling from Argentina to Cape Verde.
- **GEOGRAPHIC CONTEXT:** Endemic to the **ANDES** region, South America.
- **THE PATHOGEN:** Zoonotic, **RODENT-BORNE** virus.
- **TRANSMISSION:** Infected rodent urine, droppings, saliva.
*NOT AIRBORNE (H-to-H rare).
- **DISEASE:** **HANTAVIRUS PULMONARY SYNDROME (HPS)**.
Severe respiratory disease.
- **GLOBAL SIGNIFICANCE:** Vulnerability of maritime travel, importance of disease surveillance.



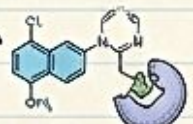
2. TRANSCRIPTFORMER: AI & CELL IDENTIFICATION

- **NEW AI TOOLS:** Cutting-edge Artificial Intelligence for cellular biology.
- **TRANSFORMER-BASED:** Uses transformer architectures, like language models.
- **TRANSCRIPTOMIC DATA:** Analyzes mRNA to identify cells and their functional states.
- **APPLICATIONS:** Single-cell analysis, complex tissue structures, personalized medicine.



3. PANCREATIC CANCER & DARAXONRASIB

- **THE PROBLEM:** **PANCREATIC CANCER**.
- **KRAS GENE MUTATION:** Normally regulates cell division; mutated leads to **UNREGULATED CELL GROWTH**.
- **THE CHALLENGE:** Often "undruggable".
- **THERAPY:** **DARAXONRASIB** (new agent).
- **MECHANISM:** Designed to **TARGET MULTIPLE** mutations in the **RAS FAMILY**. Inhibits cancer signaling pathways.



'Daraxonrasib'

4. AGNI-V & MISSION DIVYASTRA

- **MISSILE CLASS:** Intercontinental Ballistic Missile (ICBM) class (Range 5,000+ km).
- **SPECIFICATIONS:** Three-stage missile. Uses **SOLID FUEL**. Quicker launch.
- **MISSION DIVYASTRA:** The first flight test of Agni-5 with new technology.
- **MIRV TECHNOLOGY:** Multiple Independently Targetable Re-entry Vehicles. Single missile carries **MULTIPLE WARHEADS**, each hitting **DIFFERENT** targets.
- **STRATEGIC IMPORTANCE:** Ensures 'Credible Minimum Deterrence'. Strengthens 'No First Use' policy.



COMPREHENSIVE NOTEBOOK EDITION

DEFENCE: DRDO - IAF TARA GLIDE WEAPON



- ⑤ **TARA Glide Weapon (TARA)**
Tactical Advanced Range Augmentation
Maiden Flight Trial from Jaguar Fighter (May 2026)



NATURE:

- First Indigenous Glide Weapon System
- Converts Conventional 'Dumb Bombs' (unguided warheads)

FUNCTION: PRECISION STRIKE

- On-board GNC (Guidance, Navigation, & Control)
- Deployable Wings for Extended Range
- Pinpoint Accuracy

STRATEGIC ADVANTAGE:

- **STAND-OFF CAPABILITY** (outside enemy AD range)
- **COST-EFFECTIVE** (cheaper than missiles, no engine)
- Fits with SAAW, GAURAV/GAUTHAM glide bombs



POLITY: ARTICLE 164 & GOVERNOR'S POWERS



- ⑥ **GOVERNOR APPOINTS CM (Art 164)**

Article 164(1): CM appointed by Governor
- Discretionary Power (if no clear majority)



GOVERNED BY: CONSTITUTIONAL MORALITY

- Governor as Formal Head, not a parallel power

FLOOR TEST MANDATE:

- Only place to test majority is **THE FLOOR OF THE HOUSE**
- NOT the Governor's Office
- **S.R. BOMMAI vs. UOI (1994)** Precedent

RECENT CONTEXT (e.g., TN Case):

- governor cannot act as arbiter on private letters
- delay in cabinet advice violates parliamentary system



ENVIRONMENT: D'ERING MEMORIAL WILDLIFE SANCTUARY



- ⑦ **ARUNACHAL PRADESH**

Location: East Siang District (nr Pasighat)
- Nestled between **SIANG & SIBIYA RIVERS**



KEY FINDING (May 2026):

- **ROYAL BENGAL TIGER** seen via camera trap after nearly 20 years!

ECOLOGY & HABITAT:

- **UNIQUE: 80% AREA GRASSLAND** (Riverine/Alluvial)
- Dense Riverine Forest

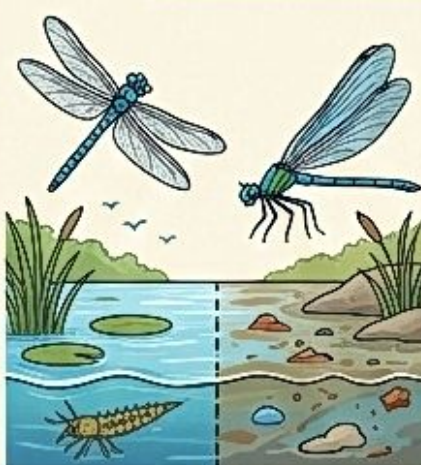


SIGNIFICANCE:

- **VITAL WILDLIFE CORRIDOR** (Assam-Arunachal)
- Habitat for **CHINESE PANGOLIN**, **HISPID HARE** (Critically Endangered)



BIODIVERSITY: ODONATES AS BIO-INDICATORS



- ⑧ **DRAGONFLIES & DAMSELFLIES MISSING**
ORDER: **ODONATA**



- Western Ghats Study: ~35% Species Missing from historical habitats

KEY CONCEPT: BIO-INDICATORS / PROXY MEASURES

- Dual Life Cycle: Aquatic Larvae + Terrestrial Adults
- Highly Sensitive to Water/Air Quality

ROLE IN ECOSYSTEM:

- Apex predators (mosquito control)
- 'PROXY' for ecosystem health, similar to **LICHENS** (Air) & **ALGAE** (Water nutrients)



CAUSE OF DECLINE:

- Linear infrastructure projects
- Changes in land-use patterns

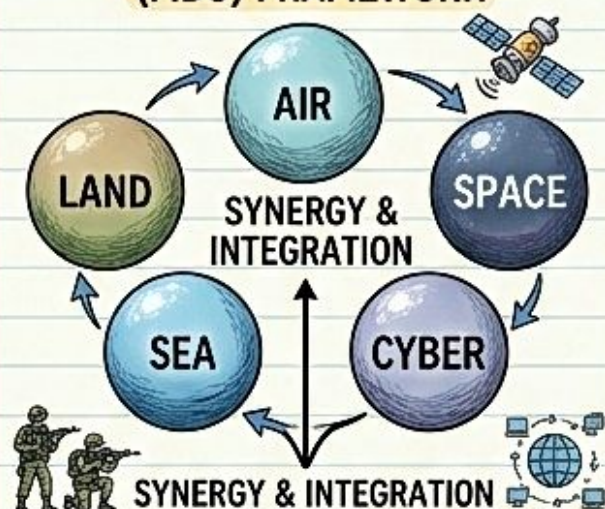


UPSC CSE 2026: CURRENT AFFAIRS STUDY NOTES - HIGH-YIELD TOPICS

(9) OPERATION SINDOOR & MULTI-DOMAIN OPERATIONS (MDO)

- **CONTEXT: MAY 2025 - INDIA'S TRANSITION TO NEXT-GEN WARFARE**

MULTI-DOMAIN OPERATIONS (MDO) FRAMEWORK



- **CORE CONCEPT: SYNERGY & STRATEGIC UNPREDICTABILITY (PRESENTING ADVERSARY WITH MULTIPLE DILEMMAS)**



- **TECHNOLOGICAL PILLARS: CYBER, AI, AUTONOMOUS SYSTEMS (SWARM DRONES, SAKSHAM)**



- **JOINT DOCTRINE FOR INTEGRATED COMMUNICATION ARCHITECTURE**

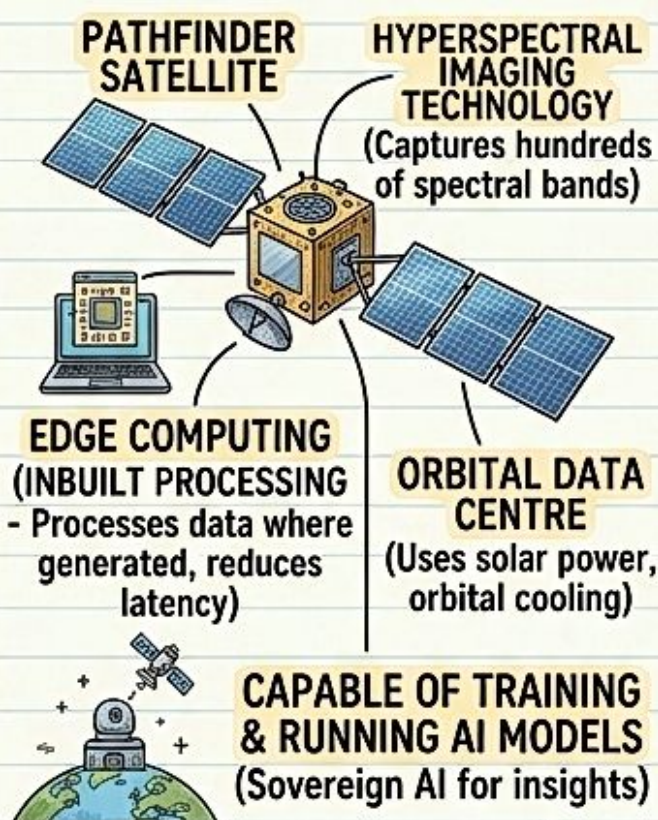
(Ensures Secure, Real-Time Navy-Army-IAF links)



- **STRATEGIC SIGNIFICANCE:** Move from 'Reactive Defense' to 'Calibrated Retribution'. Proved India can manage escalation under a nuclear shadow.

(10) INDIA'S FIRST ORBITAL DATA CENTRE (ODC) SATELLITE

- **CONTEXT: MAY 2026 - PIXXEL & SARVAM AI PARTNERSHIP ON 'PATHFINDER' MISSION**



- **WHY ODC?:**



- Latency reduction for real-time Disaster Management/Defense.
- Resource efficiency (unlimited solar power).



- **TECHNICAL DIFFICULTIES:**

- Thermal Management (Heat dissipation in vacuum).
- Radiation Hardening (Protecting GPU processors).

- **WAY FORWARD:**



- Potential constellation of ODCs making India a global hub for 'Space-as-a-Service'.

